Completeness of the minimal normal modal logic K over all models

The Open Logic Project contributors

openlogic:normal-modal-logic/completeness:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the slash-proves and slash-sat macros of the statement render the negated relations as printed; the sentence introducing the corollary ('Since the canonical model for K determines K...') sits outside the span.

Editorial note. The label cor:Kcomplete names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own cor environment in the unit source.

Allowed standard prerequisites (editorial).

- Determination theorem for the canonical model. Same part, \ollabel{thm:determination} (source line 598, own card). Line 670: converts non-provability into canonical failure of validity.
- Canonical model is a model. Construction of the canonical model K: worlds are maximal consistent sets - legitimate structures. Implicit in line 670's step from canonical failure to plain invalidity.

Corollary 0.1. *The basic modal logic K is complete with respect to the class of all models, i.e., if* $\models A$ *then* $\mathbf{K} \vdash A$.

Proof. Contrapositively, if $\mathbf{K} \vdash /A$ then by Determination $/ \models M^{\mathbf{K}}[A]$ and hence A is not valid. □